Probability — Experimental probability · Estimating probability

Give probabilities as fractions, decimals or percentages. ★ = challenge question.

1 Mia dropped a plastic cup 40 times and recorded how it landed.

Outcome	Frequency	Experimental probability
on its side	29	
upright	8	
upside down	3	

a Copy and complete the table.

b Mia predicted that the cup would be unlikely to land upright. Was she right? Explain.

c How could Mia make her estimates more reliable?

2 In an honesty experiment, 80 wallets were left in a street. 32 were handed in at a shop, 28 at a police station and 20 were kept.

a Work out the experimental probability of each outcome.

b Another wallet is left in the street. Estimate the probability that it is handed in somewhere.

3 In a video game a bonus item appears at random 3 times in every 24 levels. Work out the probability that the bonus appears in a particular level.

4 In a survey, 8% of people who set themselves a fitness target achieved it. Another survey questions 350 people with a fitness target. How many would you expect to achieve it?

5 A bakery bakes 500 loaves. 15 are burnt and thrown away, 380 are sold at full price and the rest are sold at a discount.

a Work out the relative frequency of each outcome.

b Estimate the probability that a loaf is *not* burnt.

c Next week the bakery bakes 2000 loaves. How many would you expect to be sold at a discount?

6 The grades of 300 students in an exam were: A* 60, A 90, B 75, C 45, D 20, E 10.

- **a** How many students got grade B or C?
- **b** Estimate the probability that a student next year gets grade B or C.
- **c** 150 students sit the exam next year. How many would you expect to get grade B or C?

7 A dice is rolled 60 times and lands on 6 fifteen times.

- **a** How many sixes would you expect from a fair dice in 60 rolls?
- **b** What is the experimental probability of a 6?
- **c** Does this prove the dice is biased? Explain.

8 A spinner has three sectors of different sizes: red, blue and green. In 200 spins the results were red 96, blue 64, green 40.

- **a** Estimate the probability of each colour.
- **b** Which sector is the largest? Explain.
- **c** How many greens would you expect in 1000 spins?

Check: experimental probability = frequency ÷ total trials · all the probabilities of an experiment add up to 1 · expected number = probability × trials · more trials → better estimate.